

Grade 10 - Term 3

Catch Ups Solution

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Evaluated criteria

- C5. Uses the standard normal distribution and its tables to calculate probabilities.
- C6. Uses normal distributions in contextual situations.
- C7. Uses inverse normal reasoning to find values and intervals.

1 (14) Standard Normal Table Problems (C5)

Let $Z \sim N(0, 1)$. Let $\Phi(z) = P(Z < z)$.

This section trains students to compute standard-normal probabilities using the table.

Questions/Tasks.

1) (6) Direct and complement-rule probabilities.

- Find $P(Z < 1.24)$.
- Find $P(Z > -0.67)$.
- Find $P(Z > 1.18)$.
- Find $P(Z < -0.52)$.
- Find $P(Z < 0.36)$.
- Find $P(Z > 2.05)$.
- Find $P(Z < -1.41)$.

- h) Find $P(Z > -1.09)$.
- i) Find $P(Z > 0.73)$.
- j) Find $P(Z < 1.96)$.
- k) Find $P(Z < -0.88)$.
- l) Find $P(Z > -2.17)$.

2) (4) **Interval probabilities.**

- a) Find $P(0.42 < Z < 1.37)$.
- b) Find $P(-0.95 < Z < 1.12)$.
- c) Find $P(-1.75 < Z < -0.40)$.
- d) Find $P(-2.10 < Z < 0.65)$.

3) (4) **Tail-union probabilities.**

- a) Find $P(Z < -1.25 \text{ or } Z > 1.80)$.
- b) Find $P(Z < 0.45 \text{ or } Z > 1.70)$.
- c) Find $P(Z < -2.00 \text{ or } Z > -0.60)$.
- d) Find $P(Z < -0.30 \text{ or } Z > 2.25)$.

1. (6) **Direct and complement-rule probabilities**

Find the following probabilities.

$$\text{Problem 1} \quad P(Z < 1.24) \quad P(Z > -0.67)$$

$$P(Z < 1.24) = \Phi(1.24) = 0.8925$$

$$P(Z > -0.67) = 1 - \Phi(-0.67)$$

$$\Phi(-0.67) = 1 - \Phi(0.67)$$

$$P(Z > -0.67) = 1 - (1 - \Phi(0.67)) = \Phi(0.67) = 0.7486$$

$$\text{Problem 2} \quad P(Z > 1.18) \quad P(Z < -0.52)$$

$$P(Z > 1.18) = 1 - \Phi(1.18) = 1 - 0.8810 = 0.1190$$

$$\Phi(-x) = 1 - \Phi(x)$$

$$P(Z < -0.52) = \Phi(-0.52) = 1 - \Phi(0.52) = 1 - 0.6985 = 0.3015$$

$$\textbf{Problem 3} \quad P(Z < 0.36) \quad P(Z > 2.05)$$

$$P(Z < 0.36) = \Phi(0.36) = 0.6406$$

$$P(Z > 2.05) = 1 - \Phi(2.05) = 1 - 0.9798 = 0.0202$$

$$\textbf{Problem 4} \quad P(Z < -1.41) \quad P(Z > -1.09)$$

$$P(Z < -1.41) = \Phi(-1.41) = 1 - \Phi(1.41) = 1 - 0.9207 = 0.0793$$

$$P(Z > -1.09) = 1 - \Phi(-1.09)$$

$$\Phi(-1.09) = 1 - \Phi(1.09)$$

$$P(Z > -1.09) = 1 - (1 - \Phi(1.09)) = \Phi(1.09) = 0.8621$$

$$\textbf{Problem 5} \quad P(Z > 0.73) \quad P(Z < 1.96)$$

$$P(Z > 0.73) = 1 - \Phi(0.73) = 1 - 0.7673 = 0.2327$$

$$P(Z < 1.96) = \Phi(1.96) = 0.9750$$

$$\textbf{Problem 6} \quad P(Z < -0.88) \quad P(Z > -2.17)$$

$$P(Z < -0.88) = \Phi(-0.88) = 1 - \Phi(0.88) = 1 - 0.8106 = 0.1894$$

$$P(Z > -2.17) = 1 - \Phi(-2.17)$$

$$\Phi(-2.17) = 1 - \Phi(2.17)$$

$$P(Z > -2.17) = 1 - (1 - \Phi(2.17)) = \Phi(2.17) = 0.9850$$



2. (2) Interval probabilities

Find the following interval probabilities.

$$\textbf{Problem 7} \quad P(0.42 < Z < 1.37) \quad P(-0.95 < Z < 1.12)$$

$$P(0.42 < Z < 1.37) = \Phi(1.37) - \Phi(0.42)$$

$$P(0.42 < Z < 1.37) = 0.9147 - 0.6628 = 0.2519$$

$$P(-0.95 < Z < 1.12) = \Phi(1.12) - \Phi(-0.95)$$

$$\Phi(-0.95) = 1 - \Phi(0.95) = 1 - 0.8289 = 0.1711$$

$$P(-0.95 < Z < 1.12) = 0.8686 - 0.1711 = 0.6975$$

$$\textbf{Problem 8} \quad P(-1.75 < Z < -0.40) \quad P(-2.10 < Z < 0.65)$$

$$P(-1.75 < Z < -0.40) = \Phi(-0.40) - \Phi(-1.75)$$

$$\Phi(-0.40) = 1 - \Phi(0.40) = 1 - 0.6554 = 0.3446$$

$$\Phi(-1.75) = 1 - \Phi(1.75) = 1 - 0.9599 = 0.0401$$

$$P(-1.75 < Z < -0.40) = 0.3446 - 0.0401 = 0.3045$$

$$P(-2.10 < Z < 0.65) = \Phi(0.65) - \Phi(-2.10)$$

$$\Phi(-2.10) = 1 - \Phi(2.10) = 1 - 0.9821 = 0.0179$$

$$P(-2.10 < Z < 0.65) = 0.7422 - 0.0179 = 0.7243$$



3. (2) Tail-union probabilities

Find the following tail-union probabilities.

$$\textbf{Problem 9} \quad P(Z < -1.25 \text{ or } Z > 1.80) \quad P(Z < 0.45 \text{ or } Z > 1.70)$$

$$P(Z < -1.25 \text{ or } Z > 1.80) = P(Z < -1.25) + P(Z > 1.80)$$

$$P(Z < -1.25) = \Phi(-1.25) = 1 - \Phi(1.25) = 1 - 0.8944 = 0.1056$$

$$P(Z > 1.80) = 1 - \Phi(1.80) = 1 - 0.9641 = 0.0359$$

$$P(Z < -1.25 \text{ or } Z > 1.80) = 0.1056 + 0.0359 = 0.1415$$

$$P(Z < 0.45 \text{ or } Z > 1.70) = P(Z < 0.45) + P(Z > 1.70)$$

$$P(Z < 0.45) = \Phi(0.45) = 0.6736$$

$$P(Z > 1.70) = 1 - \Phi(1.70) = 1 - 0.9554 = 0.0446$$

$$P(Z < 0.45 \text{ or } Z > 1.70) = 0.6736 + 0.0446 = 0.7182$$

$$\textbf{Problem 10} \quad P(Z < -2.00 \text{ or } Z > -0.60) \quad P(Z < -0.30 \text{ or } Z > 2.25)$$

$$P(Z < -2.00 \text{ or } Z > -0.60) = P(Z < -2.00) + P(Z > -0.60)$$

$$P(Z < -2.00) = \Phi(-2.00) = 1 - \Phi(2.00) = 1 - 0.9772 = 0.0228$$

$$P(Z > -0.60) = 1 - \Phi(-0.60)$$

$$\Phi(-0.60) = 1 - \Phi(0.60) = 1 - 0.7257 = 0.2743$$

$$P(Z > -0.60) = 1 - 0.2743 = 0.7257$$

$$P(Z < -2.00 \text{ or } Z > -0.60) = 0.0228 + 0.7257 = 0.7485$$

$$P(Z < -0.30 \text{ or } Z > 2.25) = P(Z < -0.30) + P(Z > 2.25)$$

$$P(Z < -0.30) = \Phi(-0.30) = 1 - \Phi(0.30) = 1 - 0.6179 = 0.3821$$

$$P(Z > 2.25) = 1 - \Phi(2.25) = 1 - 0.9878 = 0.0122$$

$$P(Z < -0.30 \text{ or } Z > 2.25) = 0.3821 + 0.0122 = 0.3943$$



2 (14) Contextual Normal Distribution Problems (C6)

Use

$$Z = \frac{X - \mu}{\sigma}$$

to convert each contextual value into a standard normal value.

Questions/Tasks.

- 1) (1) **Exam scores.** If exam scores follow $X \sim N(75, 10^2)$, find $P(X < 87.4)$ and $P(X > 68.3)$.
- 2) (1) **Delivery times.** If delivery times follow $X \sim N(40, 6^2)$, find $P(X > 47.08)$ and $P(X < 36.88)$.
- 3) (1) **Product weights.** If product weights follow $X \sim N(500, 20^2)$, find $P(X < 507.2)$ and $P(X > 541)$.
- 4) (1) **Commute times.** If commute times follow $X \sim N(35, 8^2)$, find $P(X < 23.72)$ and $P(X > 26.28)$.
- 5) (1) **Customer spending.** If customer spending follows $X \sim N(120, 15^2)$, find $P(X > 130.95)$ and $P(X < 149.4)$.
- 6) (1) **Machine lifetimes.** If machine lifetimes follow $X \sim N(1000, 100^2)$, find $P(X < 912)$ and $P(X > 783)$.
- 7) (2) **Weekly app usage.** If weekly app usage follows $X \sim N(20, 4^2)$, find $P(21.68 < X < 25.48)$ and $P(16.2 < X < 24.48)$.
- 8) (2) **Crop yields.** If crop yields follow $X \sim N(60, 5^2)$, find $P(51.25 < X < 58)$ and $P(49.5 < X < 63.25)$.
- 9) (2) **Battery duration.** If battery duration follows $X \sim N(12, 2^2)$, find $P(X < 9.5 \text{ or } X > 15.6)$ and $P(X < 12.9 \text{ or } X > 15.4)$.
- 10) (2) **Scholarship scores.** If scholarship scores follow $X \sim N(300, 30^2)$, find $P(X < 240 \text{ or } X > 282)$ and $P(X < 291 \text{ or } X > 367.5)$.

1. (1) Exam scores

Exam scores follow

$$X \sim N(75, 10^2)$$

First probability. Find $P(X < 87.4)$.

$$P(X < 87.4) = P\left(Z < \frac{87.4 - 75}{10}\right) = P(Z < 1.24)$$

$$P(X < 87.4) = \Phi(1.24) = 0.8925$$

Second probability. Find $P(X > 68.3)$.

$$P(X > 68.3) = P\left(Z > \frac{68.3 - 75}{10}\right) = P(Z > -0.67)$$

$$P(X > 68.3) = 1 - \Phi(-0.67) = 1 - 0.2514 = 0.7486$$

This uses the left-tail and complement-rule logic from C5.



2. (1) Delivery times

Delivery times follow

$$X \sim N(40, 6^2)$$

First probability. Find $P(X > 47.08)$.

$$P(X > 47.08) = P\left(Z > \frac{47.08 - 40}{6}\right) = P(Z > 1.18)$$

$$P(X > 47.08) = 1 - \Phi(1.18) = 1 - 0.8810 = 0.1190$$

Second probability. Find $P(X < 36.88)$.

$$P(X < 36.88) = P\left(Z < \frac{36.88 - 40}{6}\right) = P(Z < -0.52)$$

$$P(X < 36.88) = \Phi(-0.52) = 0.3015$$

This uses the complement-rule and left-tail table logic from C5.



3. (1) Product weights

Product weights follow

$$X \sim N(500, 20^2)$$

First probability. Find $P(X < 507.2)$.

$$P(X < 507.2) = P\left(Z < \frac{507.2 - 500}{20}\right) = P(Z < 0.36)$$

$$P(X < 507.2) = \Phi(0.36) = 0.6406$$

Second probability. Find $P(X > 541)$.

$$P(X > 541) = P\left(Z > \frac{541 - 500}{20}\right) = P(Z > 2.05)$$

$$P(X > 541) = 1 - \Phi(2.05) = 1 - 0.9798 = 0.0202$$

This uses the left-tail and complement-rule logic from C5.



4. (1) Commute times

Commute times follow

$$X \sim N(35, 8^2)$$

First probability. Find $P(X < 23.72)$.

$$P(X < 23.72) = P\left(Z < \frac{23.72 - 35}{8}\right) = P(Z < -1.41)$$

$$P(X < 23.72) = \Phi(-1.41) = 0.0793$$

Second probability. Find $P(X > 26.28)$.

$$P(X > 26.28) = P\left(Z > \frac{26.28 - 35}{8}\right) = P(Z > -1.09)$$

$$P(X > 26.28) = 1 - \Phi(-1.09) = 1 - 0.1379 = 0.8621$$

This uses the left-tail and complement-rule logic from C5.



5. (1) Customer spending

Customer spending follows

$$X \sim N(120, 15^2)$$

First probability. Find $P(X > 130.95)$.

$$P(X > 130.95) = P\left(Z > \frac{130.95 - 120}{15}\right) = P(Z > 0.73)$$

$$P(X > 130.95) = 1 - \Phi(0.73) = 1 - 0.7673 = 0.2327$$

Second probability. Find $P(X < 149.4)$.

$$P(X < 149.4) = P\left(Z < \frac{149.4 - 120}{15}\right) = P(Z < 1.96)$$

$$P(X < 149.4) = \Phi(1.96) = 0.9750$$

This uses the complement-rule and left-tail table logic from C5.



6. (1) Machine lifetimes

Machine lifetimes follow

$$X \sim N(1000, 100^2)$$

First probability. Find $P(X < 912)$.

$$P(X < 912) = P\left(Z < \frac{912 - 1000}{100}\right) = P(Z < -0.88)$$

$$P(X < 912) = \Phi(-0.88) = 0.1894$$

Second probability. Find $P(X > 783)$.

$$P(X > 783) = P\left(Z > \frac{783 - 1000}{100}\right) = P(Z > -2.17)$$

$$P(X > 783) = 1 - \Phi(-2.17) = 1 - 0.0150 = 0.9850$$

This uses the left-tail and complement-rule logic from C5.



7. (2) Weekly app usage

Weekly app usage follows

$$X \sim N(20, 4^2)$$

First probability. Find $P(21.68 < X < 25.48)$.

$$P(21.68 < X < 25.48) = P\left(\frac{21.68 - 20}{4} < Z < \frac{25.48 - 20}{4}\right)$$

$$P(21.68 < X < 25.48) = P(0.42 < Z < 1.37)$$

$$P(21.68 < X < 25.48) = \Phi(1.37) - \Phi(0.42) = 0.9147 - 0.6628 = 0.2519$$

Second probability. Find $P(16.2 < X < 24.48)$.

$$P(16.2 < X < 24.48) = P\left(\frac{16.2 - 20}{4} < Z < \frac{24.48 - 20}{4}\right)$$

$$P(16.2 < X < 24.48) = P(-0.95 < Z < 1.12)$$

$$P(16.2 < X < 24.48) = \Phi(1.12) - \Phi(-0.95) = 0.8686 - 0.1711 = 0.6975$$

This uses the interval logic from C5.



8. (2) Crop yields

Crop yields follow

$$X \sim N(60, 5^2)$$

First probability. Find $P(51.25 < X < 58)$.

$$P(51.25 < X < 58) = P\left(\frac{51.25 - 60}{5} < Z < \frac{58 - 60}{5}\right)$$

$$P(51.25 < X < 58) = P(-1.75 < Z < -0.40)$$

$$P(51.25 < X < 58) = \Phi(-0.40) - \Phi(-1.75) = 0.3446 - 0.0401 = 0.3045$$

Second probability. Find $P(49.5 < X < 63.25)$.

$$P(49.5 < X < 63.25) = P\left(\frac{49.5 - 60}{5} < Z < \frac{63.25 - 60}{5}\right)$$

$$P(49.5 < X < 63.25) = P(-2.10 < Z < 0.65)$$

$$P(49.5 < X < 63.25) = \Phi(0.65) - \Phi(-2.10) = 0.7422 - 0.0179 = 0.7243$$

This uses the interval logic from C5.



9. (2) Battery duration

Battery duration follows

$$X \sim N(12, 2^2)$$

First probability. Find $P(X < 9.5 \text{ or } X > 15.6)$.

$$P(X < 9.5 \text{ or } X > 15.6) = P\left(Z < \frac{9.5 - 12}{2}\right) + P\left(Z > \frac{15.6 - 12}{2}\right)$$

$$P(X < 9.5 \text{ or } X > 15.6) = P(Z < -1.25) + P(Z > 1.80)$$

$$P(X < 9.5 \text{ or } X > 15.6) = \Phi(-1.25) + 1 - \Phi(1.80)$$

$$P(X < 9.5 \text{ or } X > 15.6) = 0.1056 + 1 - 0.9641 = 0.1415$$

Second probability. Find $P(X < 12.9 \text{ or } X > 15.4)$.

$$P(X < 12.9 \text{ or } X > 15.4) = P\left(Z < \frac{12.9 - 12}{2}\right) + P\left(Z > \frac{15.4 - 12}{2}\right)$$

$$P(X < 12.9 \text{ or } X > 15.4) = P(Z < 0.45) + P(Z > 1.70)$$

$$P(X < 12.9 \text{ or } X > 15.4) = \Phi(0.45) + 1 - \Phi(1.70)$$

$$P(X < 12.9 \text{ or } X > 15.4) = 0.6736 + 1 - 0.9554 = 0.7182$$

This uses the tail-union logic from C5.



10. (2) Scholarship scores

Scholarship scores follow

$$X \sim N(300, 30^2)$$

First probability. Find $P(X < 240 \text{ or } X > 282)$.

$$P(X < 240 \text{ or } X > 282) = P\left(Z < \frac{240 - 300}{30}\right) + P\left(Z > \frac{282 - 300}{30}\right)$$

$$P(X < 240 \text{ or } X > 282) = P(Z < -2.00) + P(Z > -0.60)$$

$$P(X < 240 \text{ or } X > 282) = \Phi(-2.00) + 1 - \Phi(-0.60)$$

$$P(X < 240 \text{ or } X > 282) = 0.0228 + 1 - 0.2743 = 0.7485$$

Second probability. Find $P(X < 291 \text{ or } X > 367.5)$.

$$P(X < 291 \text{ or } X > 367.5) = P\left(Z < \frac{291 - 300}{30}\right) + P\left(Z > \frac{367.5 - 300}{30}\right)$$

$$P(X < 291 \text{ or } X > 367.5) = P(Z < -0.30) + P(Z > 2.25)$$

$$P(X < 291 \text{ or } X > 367.5) = \Phi(-0.30) + 1 - \Phi(2.25)$$

$$P(X < 291 \text{ or } X > 367.5) = 0.3821 + 1 - 0.9878 = 0.3943$$

This uses the tail-union logic from C5.



3 (14) Inverse Normal Problems (C7)

Use

$$Z = \frac{X - \mu}{\sigma} \quad \text{and} \quad x = \mu + \sigma z$$

to convert between standard normal values and contextual values.

Questions/Tasks.

1) (1) **Commute times.** Commute times follow $X \sim N(38, 7^2)$.

a) Find x such that $P(X < x) = 0.80$.

b) Find x such that $P(X > x) = 0.10$.

2) (1) **Exam scores.** Exam scores follow $X \sim N(72, 9^2)$.

a) Find x such that $P(X > x) = 0.80$.

b) Find x such that $P(X < x) = 0.25$.

3) (1) **Delivery times.** Delivery times follow $X \sim N(56, 5^2)$.

a) Find x such that $P(X < x) = 0.60$.

b) Find x such that $P(X > x) = 0.30$.

4) **(1) Product weights.** Product weights follow $X \sim N(500, 20^2)$.

- a) Find x such that $P(X < x) = 0.10$.
- b) Find x such that $P(X > x) = 0.75$.

5) **(1) Customer spending.** Customer spending follows $X \sim N(120, 15^2)$.

- a) Find x such that $P(X > x) = 0.05$.
- b) Find x such that $P(X < x) = 0.70$.

6) **(1) App usage.** App usage follows $X \sim N(20, 4^2)$.

- a) Find x such that $P(X < x) = 0.40$.
- b) Find x such that $P(X > x) = 0.90$.

7) **(2) Battery life.** Battery life follows $X \sim N(20, 2.5^2)$.

- a) Find $[a, b]$ such that $P(a < X < b) = 0.90$, with a central interval.
- b) Find $[a, b]$ such that $P(X < a) = 0.60$ and $P(X < b) = 0.90$.

8) **(2) Machine lifetimes.** Machine lifetimes follow $X \sim N(1000, 100^2)$.

- a) Find $[a, b]$ such that $P(X < a) = 0.10$ and $P(X < b) = 0.40$.
- b) Find $[a, b]$ such that $P(X < a) = 0.25$ and $P(X < b) = 0.75$.

9) **(2) Metal rod diameters.** Metal rod diameters follow $X \sim N(10.00, 0.08^2)$.

- a) Find a and b such that $P(X < a \text{ or } X > b) = 0.10$, with symmetric tails.
- b) Find a and b such that $P(X < a \text{ or } X > b) = 0.70$, with $P(X < a) = 0.60$ and $P(X > b) = 0.10$.

10) **(2) Crop yields.** Crop yields follow $X \sim N(60, 5^2)$.

- a) Find a and b such that $P(X < a \text{ or } X > b) = 0.70$, with $P(X < a) = 0.10$ and $P(X > b) = 0.60$.
- b) Find a and b such that $P(X < a \text{ or } X > b) = 0.45$, with $P(X < a) = 0.20$ and $P(X > b) = 0.25$.

1. (1) Commute times

Commute times follow

$$X \sim N(38, 7^2)$$

First cutoff. Find x such that $P(X < x) = 0.80$.

$$P(X < x) = 0.80 \quad \longrightarrow \quad \Phi(z) = 0.80$$

$$z = 0.842$$

$$x = \mu + \sigma z = 38 + 7(0.842) = 43.894$$

$$x \approx 43.89$$

Second cutoff. Find x such that $P(X > x) = 0.10$.

$$\begin{aligned}P(X > x) = 0.10 &\longrightarrow P(X < x) = 0.90 \\ \Phi(z) = 0.90 &\longrightarrow z = 1.282 \\ x = \mu + \sigma z = 38 + 7(1.282) &= 46.974 \\ x &\approx 46.97\end{aligned}$$

This inverse step depends on the same left-tail and complement-rule interpretations practiced in C5.



2. (1) Exam scores

Exam scores follow

$$X \sim N(72, 9^2)$$

First cutoff. Find x such that $P(X > x) = 0.80$.

$$\begin{aligned}P(X > x) = 0.80 &\longrightarrow P(X < x) = 0.20 \\ \Phi(z) = 0.20 &\longrightarrow z = -0.842 \\ x = \mu + \sigma z = 72 + 9(-0.842) &= 64.422 \\ x &\approx 64.42\end{aligned}$$

Second cutoff. Find x such that $P(X < x) = 0.25$.

$$\begin{aligned}P(X < x) = 0.25 &\longrightarrow \Phi(z) = 0.25 \\ z &= -0.674 \\ x = \mu + \sigma z = 72 + 9(-0.674) &= 65.934 \\ x &\approx 65.93\end{aligned}$$

This inverse step depends on the same complement-rule and left-tail probability interpretations practiced in C5.



3. (1) Delivery times

Delivery times follow

$$X \sim N(56, 5^2)$$

First cutoff. Find x such that $P(X < x) = 0.60$.

$$\begin{aligned}P(X < x) = 0.60 &\longrightarrow \Phi(z) = 0.60 \\ z &= 0.253\end{aligned}$$

$$x = \mu + \sigma z = 56 + 5(0.253) = 57.265$$

$$x \approx 57.27$$

Second cutoff. Find x such that $P(X > x) = 0.30$.

$$P(X > x) = 0.30 \quad \longrightarrow \quad P(X < x) = 0.70$$

$$\Phi(z) = 0.70 \quad \longrightarrow \quad z = 0.524$$

$$x = \mu + \sigma z = 56 + 5(0.524) = 58.620$$

$$x \approx 58.62$$

This inverse step depends on the same left-tail and complement-rule interpretations practiced in C5.



4. (1) Product weights

Product weights follow

$$X \sim N(500, 20^2)$$

First cutoff. Find x such that $P(X < x) = 0.10$.

$$P(X < x) = 0.10 \quad \longrightarrow \quad \Phi(z) = 0.10$$

$$z = -1.282$$

$$x = \mu + \sigma z = 500 + 20(-1.282) = 474.360$$

$$x \approx 474.36$$

Second cutoff. Find x such that $P(X > x) = 0.75$.

$$P(X > x) = 0.75 \quad \longrightarrow \quad P(X < x) = 0.25$$

$$\Phi(z) = 0.25 \quad \longrightarrow \quad z = -0.674$$

$$x = \mu + \sigma z = 500 + 20(-0.674) = 486.520$$

$$x \approx 486.52$$

This inverse step depends on the same left-tail and complement-rule interpretations practiced in C5.



5. (1) Customer spending

Customer spending follow

$$X \sim N(120, 15^2)$$

First cutoff. Find x such that $P(X > x) = 0.05$.

$$P(X > x) = 0.05 \quad \longrightarrow \quad P(X < x) = 0.95$$

$$\Phi(z) = 0.95 \quad \longrightarrow \quad z = 1.645$$

$$x = \mu + \sigma z = 120 + 15(1.645) = 144.675$$

$$x \approx 144.68$$

Second cutoff. Find x such that $P(X < x) = 0.70$.

$$P(X < x) = 0.70 \quad \longrightarrow \quad \Phi(z) = 0.70$$

$$z = 0.524$$

$$x = \mu + \sigma z = 120 + 15(0.524) = 127.860$$

$$x \approx 127.86$$

This inverse step depends on the same complement-rule and left-tail probability interpretations practiced in C5.



6. (1) App usage

App usage follows

$$X \sim N(20, 4^2)$$

First cutoff. Find x such that $P(X < x) = 0.40$.

$$P(X < x) = 0.40 \quad \longrightarrow \quad \Phi(z) = 0.40$$

$$z = -0.253$$

$$x = \mu + \sigma z = 20 + 4(-0.253) = 18.988$$

$$x \approx 18.99$$

Second cutoff. Find x such that $P(X > x) = 0.90$.

$$P(X > x) = 0.90 \quad \longrightarrow \quad P(X < x) = 0.10$$

$$\Phi(z) = 0.10 \quad \longrightarrow \quad z = -1.282$$

$$x = \mu + \sigma z = 20 + 4(-1.282) = 14.872$$

$$x \approx 14.87$$

This inverse step depends on the same left-tail and complement-rule interpretations practiced in C5.



7. (2) Battery life

Battery life follows

$$X \sim N(20, 2.5^2)$$

First interval cutoff. Find $[a, b]$ such that $P(a < X < b) = 0.90$, with a central interval.

$$1 - 0.90 = 0.10 \quad \longrightarrow \quad \frac{0.10}{2} = 0.05$$

$$P(X < a) = 0.05 \quad P(X < b) = 0.95$$

$$z_a = -1.645 \quad z_b = 1.645$$

$$a = \mu + \sigma z_a = 20 + 2.5(-1.645) = 15.8875$$

$$b = \mu + \sigma z_b = 20 + 2.5(1.645) = 24.1125$$

$$[a, b] \approx [15.89, 24.11]$$

Second interval cutoff. Find $[a, b]$ such that $P(X < a) = 0.60$ and $P(X < b) = 0.90$.

$$P(X < a) = 0.60 \quad P(X < b) = 0.90$$

$$z_a = 0.253 \quad z_b = 1.282$$

$$a = \mu + \sigma z_a = 20 + 2.5(0.253) = 20.6325$$

$$b = \mu + \sigma z_b = 20 + 2.5(1.282) = 23.205$$

$$[a, b] \approx [20.63, 23.21]$$

This interval cutoff uses the same interval and tail-area reasoning practiced in C5.



8. (2) Machine lifetimes

Machine lifetimes follow

$$X \sim N(1000, 100^2)$$

First interval cutoff. Find $[a, b]$ such that $P(X < a) = 0.10$ and $P(X < b) = 0.40$.

$$P(X < a) = 0.10 \quad P(X < b) = 0.40$$

$$z_a = -1.282 \quad z_b = -0.253$$

$$a = \mu + \sigma z_a = 1000 + 100(-1.282) = 871.800$$

$$b = \mu + \sigma z_b = 1000 + 100(-0.253) = 974.700$$

$$[a, b] \approx [871.80, 974.70]$$

Second interval cutoff. Find $[a, b]$ such that $P(X < a) = 0.25$ and $P(X < b) = 0.75$.

$$P(X < a) = 0.25 \quad P(X < b) = 0.75$$

$$\begin{aligned}
z_a &= -0.674 & z_b &= 0.674 \\
a &= \mu + \sigma z_a = 1000 + 100(-0.674) = 932.600 \\
b &= \mu + \sigma z_b = 1000 + 100(0.674) = 1067.400 \\
[a, b] &\approx [932.60, 1067.40]
\end{aligned}$$

This interval cutoff uses the same interval and tail-area reasoning practiced in C5.



9. (2) Metal rod diameters

Metal rod diameters follow

$$X \sim N(10.00, 0.08^2)$$

First outside-interval cutoff. Find a and b such that $P(X < a \text{ or } X > b) = 0.10$, with symmetric tails.

$$\begin{aligned}
P(X < a \text{ or } X > b) &= 0.10 & \longrightarrow & \quad \frac{0.10}{2} = 0.05 \\
P(X < a) &= 0.05 & P(X < b) &= 0.95 \\
z_a &= -1.645 & z_b &= 1.645 \\
a &= \mu + \sigma z_a = 10.00 + 0.08(-1.645) = 9.8684 \\
b &= \mu + \sigma z_b = 10.00 + 0.08(1.645) = 10.1316 \\
[a, b] &\approx [9.8684, 10.1316]
\end{aligned}$$

Second outside-interval cutoff. Find a and b such that $P(X < a \text{ or } X > b) = 0.70$, with $P(X < a) = 0.60$ and $P(X > b) = 0.10$.

$$\begin{aligned}
P(X < a) &= 0.60 & P(X > b) &= 0.10 \\
P(X < b) &= 1 - 0.10 = 0.90 \\
z_a &= 0.253 & z_b &= 1.282 \\
a &= \mu + \sigma z_a = 10.00 + 0.08(0.253) = 10.02024 \\
b &= \mu + \sigma z_b = 10.00 + 0.08(1.282) = 10.10256 \\
[a, b] &\approx [10.0202, 10.1026]
\end{aligned}$$

This outside-interval cutoff uses the same tail-area reasoning practiced in C5.



10. (2) Crop yields

Crop yields follow

$$X \sim N(60, 5^2)$$

First outside-interval cutoff. Find a and b such that $P(X < a \text{ or } X > b) = 0.70$, with $P(X < a) = 0.10$ and $P(X > b) = 0.60$.

$$P(X < a) = 0.10 \quad P(X > b) = 0.60$$

$$P(X < b) = 1 - 0.60 = 0.40$$

$$z_a = -1.282 \quad z_b = -0.253$$

$$a = \mu + \sigma z_a = 60 + 5(-1.282) = 53.590$$

$$b = \mu + \sigma z_b = 60 + 5(-0.253) = 58.735$$

$$[a, b] \approx [53.59, 58.74]$$

Second outside-interval cutoff. Find a and b such that $P(X < a \text{ or } X > b) = 0.45$, with $P(X < a) = 0.20$ and $P(X > b) = 0.25$.

$$P(X < a) = 0.20 \quad P(X > b) = 0.25$$

$$P(X < b) = 1 - 0.25 = 0.75$$

$$z_a = -0.842 \quad z_b = 0.674$$

$$a = \mu + \sigma z_a = 60 + 5(-0.842) = 55.790$$

$$b = \mu + \sigma z_b = 60 + 5(0.674) = 63.370$$

$$[a, b] \approx [55.79, 63.37]$$

This outside-interval cutoff uses the same tail-area reasoning practiced in C5.



4 C5 C6 C7 Minimal solutions

4.1 C5

Problem 1

$$P(Z < 1.24) = 0.8925$$

$$P(Z > -0.67) = 0.7486$$

Problem 2

$$P(Z > 1.18) = 0.1190$$

$$P(Z < -0.52) = 0.3015$$

Problem 3

$$P(Z < 0.36) = 0.6406$$

$$P(Z > 2.05) = 0.0202$$

Problem 4

$$P(Z < -1.41) = 0.0793$$

$$P(Z > -1.09) = 0.8621$$

Problem 5

$$P(Z > 0.73) = 0.2327$$

$$P(Z < 1.96) = 0.9750$$

Problem 6

$$P(Z < -0.88) = 0.1894$$

$$P(Z > -2.17) = 0.9850$$

Problem 7

$$P(0.42 < Z < 1.37) = 0.2519$$

$$P(-0.95 < Z < 1.12) = 0.6975$$

Problem 8

$$P(-1.75 < Z < -0.40) = 0.3045$$

$$P(-2.10 < Z < 0.65) = 0.7243$$

Problem 9

$$P(Z < -1.25 \text{ or } Z > 1.80) = 0.1415$$

$$P(Z < 0.45 \text{ or } Z > 1.70) = 0.7182$$

Problem 10

$$P(Z < -2.00 \text{ or } Z > -0.60) = 0.7485$$

$$P(Z < -0.30 \text{ or } Z > 2.25) = 0.3943$$

4.2 C6

Problem 1

$$X \sim N(75, 10^2) : \quad P(X < 87.4) = 0.8925$$

$$X \sim N(75, 10^2) : \quad P(X > 68.3) = 0.7486$$

Problem 2

$$X \sim N(40, 6^2) : \quad P(X > 47.08) = 0.1190$$

$$X \sim N(40, 6^2) : \quad P(X < 36.88) = 0.3015$$

Problem 3

$$X \sim N(500, 20^2) : \quad P(X < 507.2) = 0.6406$$

$$X \sim N(500, 20^2) : \quad P(X > 541) = 0.0202$$

Problem 4

$$X \sim N(35, 8^2) : \quad P(X < 23.72) = 0.0793$$

$$X \sim N(35, 8^2) : \quad P(X > 26.28) = 0.8621$$

Problem 5

$$X \sim N(120, 15^2) : \quad P(X > 130.95) = 0.2327$$

$$X \sim N(120, 15^2) : \quad P(X < 149.4) = 0.9750$$

Problem 6

$$X \sim N(1000, 100^2) : \quad P(X < 912) = 0.1894$$

$$X \sim N(1000, 100^2) : \quad P(X > 783) = 0.9850$$

Problem 7

$$X \sim N(20, 4^2) : \quad P(21.68 < X < 25.48) = 0.2519$$

$$X \sim N(20, 4^2) : \quad P(16.2 < X < 24.48) = 0.6975$$

Problem 8

$$X \sim N(60, 5^2) : \quad P(51.25 < X < 58) = 0.3045$$

$$X \sim N(60, 5^2) : \quad P(49.5 < X < 63.25) = 0.7243$$

Problem 9

$$X \sim N(12, 2^2) : \quad P(X < 9.5 \text{ or } X > 15.6) = 0.1415$$

$$X \sim N(12, 2^2) : \quad P(X < 12.9 \text{ or } X > 15.4) = 0.7182$$

Problem 10

$$X \sim N(300, 30^2) : \quad P(X < 240 \text{ or } X > 282) = 0.7485$$

$$X \sim N(300, 30^2) : \quad P(X < 291 \text{ or } X > 367.5) = 0.3943$$

4.3 C7

Problem 1

$$X \sim N(38, 7^2) : P(X < x) = 0.80 \longrightarrow x \approx 43.89$$

$$X \sim N(38, 7^2) : P(X > x) = 0.10 \longrightarrow x \approx 46.97$$

Problem 2

$$X \sim N(72, 9^2) : P(X > x) = 0.80 \longrightarrow x \approx 64.42$$

$$X \sim N(72, 9^2) : P(X < x) = 0.25 \longrightarrow x \approx 65.93$$

Problem 3

$$X \sim N(56, 5^2) : P(X < x) = 0.60 \longrightarrow x \approx 57.27$$

$$X \sim N(56, 5^2) : P(X > x) = 0.30 \longrightarrow x \approx 58.62$$

Problem 4

$$X \sim N(500, 20^2) : P(X < x) = 0.10 \longrightarrow x \approx 474.36$$

$$X \sim N(500, 20^2) : P(X > x) = 0.75 \longrightarrow x \approx 486.52$$

Problem 5

$$X \sim N(120, 15^2) : P(X > x) = 0.05 \longrightarrow x \approx 144.68$$

$$X \sim N(120, 15^2) : P(X < x) = 0.70 \longrightarrow x \approx 127.86$$

Problem 6

$$X \sim N(20, 4^2) : P(X < x) = 0.40 \longrightarrow x \approx 18.99$$

$$X \sim N(20, 4^2) : P(X > x) = 0.90 \longrightarrow x \approx 14.87$$

Problem 7

$$X \sim N(20, 2.5^2) : P(a < X < b) = 0.90 \longrightarrow a \approx 15.89, \quad b \approx 24.11$$

$$X \sim N(20, 2.5^2) : P(X < a) = 0.60, \quad P(X < b) = 0.90 \longrightarrow a \approx 20.63, \quad b \approx 23.21$$

Problem 8

$$X \sim N(1000, 100^2) : P(X < a) = 0.10, \quad P(X < b) = 0.40 \longrightarrow a \approx 871.80, \quad b \approx 974.70$$

$$X \sim N(1000, 100^2) : P(X < a) = 0.25, \quad P(X < b) = 0.75 \longrightarrow a \approx 932.60, \quad b \approx 1067.40$$

Problem 9

$$X \sim N(10.00, 0.08^2) : P(X < a \text{ or } X > b) = 0.10 \longrightarrow a \approx 9.8684, \quad b \approx 10.1316$$

$$X \sim N(10.00, 0.08^2) : P(X < a \text{ or } X > b) = 0.70 \longrightarrow a \approx 10.0202, \quad b \approx 10.1026$$

Problem 10

$$X \sim N(60, 5^2) : \quad P(X < a \text{ or } X > b) = 0.70 \longrightarrow a \approx 53.59, \quad b \approx 58.74$$

$$X \sim N(60, 5^2) : \quad P(X < a \text{ or } X > b) = 0.45 \longrightarrow a \approx 55.79, \quad b \approx 63.37$$